গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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Ref. no: CRTC/CEE/G20221090/S – 198

SUB-SOIL INVESTIGATION REPORT

FOR THE PROPOSED CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING

AT BOTESWAR, SYLHET

DAG # 209, KHA # 582, J.L # 66, MOUJA- CHUABAHAR BOTESWAR

CLIENT

HAZI MD. TAYFUL HAQUE
S/O – HARIS ALI

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

Verification Code: Edmu3wrS
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SAMPLE COLLECTED BY

THE SOIL ENGINEERS & CONSULTANT
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

Objectives

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Bore hole no	Depth	Quantities
01	Disturbed sample	01	0-25 ft	05
02	Disturbed sample	02	0-40 ft	08
03	Disturbed sample	03	0-40 ft	08
04	-	-	-	-
05	-	-	-	-
06	-	-	-	-
07	-	-	-	-
08	-	-	-	-
09	-	-	-	-
10	-	-	-	-

Laboratory Classifications

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a=Allowable bearing capacity, q_u= Unconfined Compressive Strength, γ= Unit Weight of soil, D= Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult}= Ultimate bearing capacity in kPa, N = SPT Number, FS= Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation=9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10** and **Table 11** respectively.

Table 7: The bearing capacity for shallow foundation in Tsf (FS=2.5)

Bore Hole No.	**Depth**	**Design N**	**Clay**						**Sand**
			Continuous			**Square (Isolated)**			
	(ft)	**-**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	**D/B=0.25**	**D/B=0.5**	**D/B=1**	
BH01	**2.5-5.0**	**24**	2.738	2.879	3.139	3.233	3.493	3.800	2.950
	5.0-10	**36**	4.132	4.346	4.738	4.880	5.272	5.735	4.453
BH02	**2.5-5.0**	**07**	0.789	0.830	0.904	0.932	1.006	1.095	0.850
	5.0-10	**15**	1.712	1.801	1.963	2.022	2.184	2.376	1.845
BH03	**2.5-5.0**	**12**	1.438	1.513	1.649	1.699	1.835	1.996	1.550
	5.0-10	**15**	1.764	1.855	2.023	2.083	2.251	2.448	1.901
BH04	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
BH05	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
BH06	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
BH07	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
BH08	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
BH09	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
BH10	**2.5-5.0**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**
	5.0-10	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

**** Design Engineer should consider the settlement issue along with the tabulated bearing capacity for designing shallow foundation**

Table 8: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth (ft)	Bore Hole									
	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.138	0.049	0.096	-	-	-	-	-	-	-
10	0.216	0.261	0.196	-	-	-	-	-	-	-
15	0.232	0.096	0.160	-	-	-	-	-	-	-
20	0.669	0.224	0.128	-	-	-	-	-	-	-
25	0.669	0.138	0.064	-	-	-	-	-	-	-
30	-	0.261	0.112	-	-	-	-	-	-	-
35	-	0.575	0.535	-	-	-	-	-	-	-
40	-	0.669	0.669	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9: The allowable end bearing for Precast driven piles in Tsf (FS=2.5)

Depth (ft)	Bore Hole									
	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	1.350	0.450	0.900	-	-	-	-	-	-	-
10	43.200	2.700	2.025	-	-	-	-	-	-	-
15	46.400	0.900	32.000	-	-	-	-	-	-	-
20	11.250	44.800	25.600	-	-	-	-	-	-	-
25	11.250	1.350	12.800	-	-	-	-	-	-	-
30	-	2.700	22.400	-	-	-	-	-	-	-
35	-	9.675	9.000	-	-	-	-	-	-	-
40	-	11.250	11.250	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)

Depth (ft)	Bore Hole									
	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.090	0.032	0.062	-	-	-	-	-	-	-
10	0.140	0.170	0.127	-	-	-	-	-	-	-
15	0.151	0.062	0.104	-	-	-	-	-	-	-
20	0.435	0.146	0.083	-	-	-	-	-	-	-
25	0.435	0.090	0.042	-	-	-	-	-	-	-
30	-	0.170	0.073	-	-	-	-	-	-	-
35	-	0.374	0.348	-	-	-	-	-	-	-
40	-	0.435	0.435	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)

Depth (ft)	Bore Hole									
	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.608	0.203	0.405	-	-	-	-	-	-	-
10	19.440	1.215	0.911	-	-	-	-	-	-	-
15	20.880	0.405	14.400	-	-	-	-	-	-	-
20	5.063	20.160	11.520	-	-	-	-	-	-	-
25	5.063	0.608	5.760	-	-	-	-	-	-	-
30	-	1.215	10.080	-	-	-	-	-	-	-
35	-	4.354	4.050	-	-	-	-	-	-	-
40	-	5.063	5.063	-	-	-	-	-	-	-
45	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
55	-	-	-	-	-	-	-	-	-	-
60	-	-	-	-	-	-	-	-	-	-
65	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

Based on the analysis and information provided by the owner following recommendations are suggested for the proposed 5-storied residential building.

Soil profile scenario

Overall soil information of the investigated site is very irregular in between the boreholes and vertical inconsistency exists in the soil formation.

In borehole 1, grey soft clay with poor bearing capacity exists at the top 5 ft depth from EGL followed by dense medium sand and hard clay layer.

In borehole 2, soft clay and sandy clay layer with poor bearing capacity exist up to 15 ft depth. Then, a dense medium sand layer has been found at 20 ft depth. After that, a 5 ft layer of soft clay and stiff has been observed at 25 ft and 30 ft depth respectively. Finally, hard clay initiates from 35 ft and continues to a greater depth.

In borehole 3, soft clay and sandy clay layer exist up to 10 ft depth from EGL. Then medium dense sand has been found up to 30 ft depth. After that, the hard clay layer initiates and continues to a greater depth.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed 6-storied residential building. Shallow foundation shall not be considered in this case. Although, bearing capacity values for designing shallow foundations can be found in the attached **Table 7** in the report which is based on existing soil conditions.

Moreover, deep foundation (Bored/precast pile) may be selected to ensure the stability of the structure. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found in the attached **Table (8 - 11)** in the report which is based on existing soil conditions.

Safety provisions

It is strongly recommended to check the **settlement** of the site considering the building load while designing either a shallow or deep foundation. Obviously, **pile load test** covering the entire project area must be carried out for confirming the design parameters of the service pile (*if pile foundation is considered*). Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be considered for the safe design of a shallow or deep foundation.

Attachments

The SPT values and site layout are shown in the Appendix. The client performed no special laboratory test for the proposed site.

Special notes

These suggestions are tentative, and the selection of type, depth, and dimensions of the foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of 25-40 ft as the client provided.

Test Assisted by:	Report Prepared by:	Countersigned by:
A H M Humayun Kabir Senior Technical Officer Dept. of CEE, SUST, Sylhet-3114	**Mohaiminul Haque** Assistant Professor Dept. of CEE, SUST, Sylhet-3114	**Prof. Dr. H M A Mahzuz** Coordinator, CRTC Dept. of CEE, SUST, Sylhet-3114

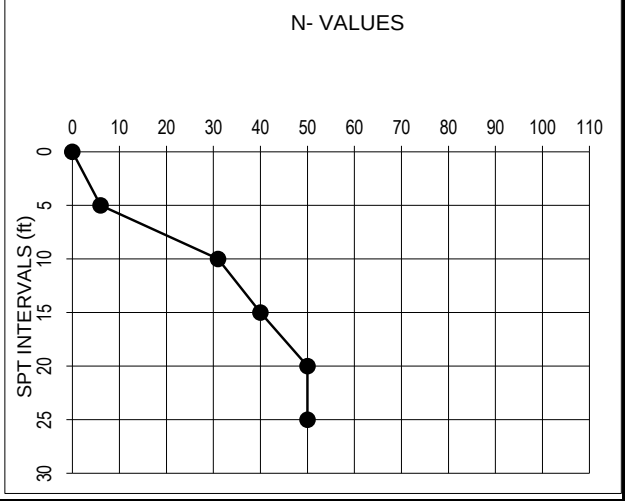
APPENDIX

RECORD OF BORING AND TESTING

CLIENT : HAZI MD. TAYFUL HAQUE	LOCATION: Boteswar, Sylhet.		Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 3'-0" BELOW EGL.		
BORE HOLE NO. **01**			Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 16/09/2022		
BORING DIA : **4 inch**			Sand with silt & clay.
DEPTH OF BORING : **25.0 ft**	COMPLETION DATE : 16/09/2022		
			Organic/Coarse sand/Gravel

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value
						0	0	0	0	0
D1		5.0	5.0	Grey soft clay		5.0	2	3	3	6
D2		10.0	10.0	Grey dense medium sand		10.0	7	12	19	31
D3		15.0				15.0	9	17	23	40
D4		20.0	10.0	Grey hard clay		20.0	14	22	28	50
D5		25.0				25.0	17	24	26	50

GRAPHICAL REPRESANTATION OF N- VALUES

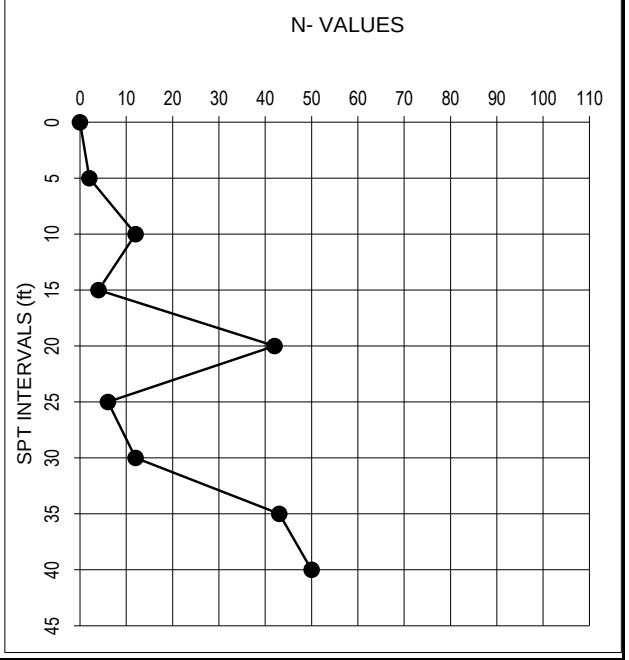


RECORD OF BORING AND TESTING

CLIENT : HAZI MD. TAYFUL HAQUE	LOCATION: Boteswar, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 3'-0" BELOW EGL.	
BORE HOLE NO. **02**		Clay with silt & sand.
METHOD OF BORING : **WASH.**	STARTING DATE : 16/09/2022	
BORING DIA : **4 inch**		Sand with silt & clay.
DEPTH OF BORING : **40.0 ft**	COMPLETION DATE : 16/09/2022	
		Organic/Coarse sand/Gravel

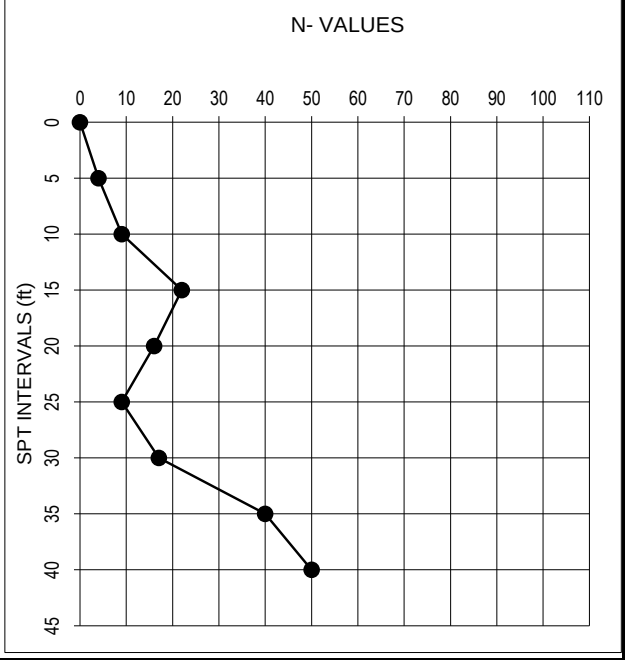
Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value
						0	0	0	0	0
D1		5.0	5.0	Grey soft clay		5.0	1	1	1	2
D2		10.0	5.0	Grey sandy clay		10.0	3	5	7	12
D3		15.0	5.0	Grey soft clay		15.0	6	2	2	4
D4		20.0	5.0	Grey dense medium sand		20.0	10	17	25	42
D5		25.0	5.0	Grey soft clay		25.0	2	3	3	6
D6		30.0	5.0	Grey stiff clay		30.0	4	5	7	12
D7		35.0	10.0	Grey hard clay		35.0	8	17	26	43
D8		40.0				40.0	15	22	28	50

GRAPHICAL REPRESANTATION OF N- VALUES



RECORD OF BORING AND TESTING

CLIENT : HAZI MD. TAYFUL HAQUE	LOCATION: Boteswar, Sylhet.	Sandy clay with silt.
PROJECT NAME : **CONSTRUCTION OF 05 STORIED RESIDENTIAL BUILDING.**	GROUND WATER LEVEL : 3'-0" BELOW EGL.	
BORE HOLE NO. **03**		Clay with silt & sand.
METHOD OF BORING : **WASH**	STARTING DATE : 16/09/2022	
BORING DIA : **4 inch**		Sand with silt & clay.
DEPTH OF BORING : **40.0 ft**	COMPLETION DATE : 16/09/2022	
		Organic/Coarse sand/Gravel

Sample No.	Type of sample	Depth Below GL(ft)	Thickness(ft)	DESCRIPTION OF SOIL STRATA	SYMBOL	SPT Intervals(ft)	06 inch	06 inch	06 inch	N- Value
						0	0	0	0	0
D1		5.0	5.0	Grey soft clay		5.0	2	2	2	4
D2		10.0	5.0	Grey sandy clay		10.0	3	4	5	9
D3		15.0	20.0	Grey medium sand with clay		15.0	7	10	12	22
D4		20.0				20.0	7	10	6	16
D5		25.0				25.0	3	4	5	9
D6		30.0				30.0	5	8	9	17
D7		35.0	10.0	Grey hard clay		35.0	10	18	22	40
D8		40.0				40.0	17	22	28	50

GRAPHICAL REPRESANTATION OF N- VALUES